PH 716 Applied Survival Analysis

Part 5: Cox Proportional Hazards Model

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What do they mean by "proportional hazards"?

- Recall the Weibull regression: $\ln T_i = \beta_0 + \sum_{j=1}^p x_{ij}\beta_j + \sigma\varepsilon_i$ with $\varepsilon_i \overset{\text{iid}}{\sim} F_{\varepsilon_i}(\epsilon) = 1 - \exp(-\exp\epsilon)$
  - $\Rightarrow \lambda_{T_i}(t) = (1/\sigma)t^{1/\sigma-1}\exp\{(-\beta_0 - \sum_{j=1}^p x_{ij}\beta_j)/\sigma\}$
  - Equiv. $\lambda_{T_i}(t) = \lambda_0(t)\exp(\sum_{j=1}^p x_{ij}\beta_j^*)$ in which $\lambda_0(t) = (1/\sigma)t^{1/\sigma-1}\exp(-\beta_0/\sigma)$ and $\beta_j^* = -\beta_j/\sigma$, $j = 1, \ldots, p$
- Under the Weibull regression, the hazard ratio (HR) between any two individuals i_1 and i_2

$$\frac{\lambda_{T_{i_1}}(t)}{\lambda_{T_{i_2}}(t)} = \exp\left(\sum_{j=1}^p x_{i_1 j}\beta_j^* - \sum_{j=1}^p x_{i_2 j}\beta_j^*\right)$$

  is constant over time
  - Hence the name "proportional hazards"
- The Weibull regression model is the only continuous-time AFT model satisfying the PH assumption

Assumptions for Cox PH model

- Observed $\widetilde{T}_i = \tilde{t}_i$ and $\Delta_i = \delta_i$ (event indicator)
- T_i are independent across i, given all covariates $x_{i1}, \ldots, x_{ip}$
- The independent and non-informative censoring
- $\lambda_{T_i}(t) = \lambda(t \mid x_{i1}, \ldots, x_{ip}) = \lambda_0(t)\exp(\sum_{j=1}^p x_{ij}\beta_j)$, or equiv.

$$\ln\lambda_{T_i}(t) = \ln\lambda_0(t) + \sum_{j=1}^p x_{ij}\beta_j$$

  - $\lambda_0(t)$ (the baseline hazard): positive for all t, obtained when all covariates are zeros and left unspecified
  - PH assumption: the HR between any two individuals, say $\lambda_{T_{i_1}}(t)/\lambda_{T_{i_2}}(t) = \exp(\sum_{j=1}^p x_{i_1 j}\beta_j - \sum_{j=1}^p x_{i_2 j}\beta_j)$, is constant over time
  - No intercept β_0 (because it is absorbed into $\lambda_0(t)$)
  - Interpretation of β_j: $\exp(\beta_j)$ is the HR associated with one-unit increase of the jth covariate, fixing everything else

Model fitting (likelihood-based)

- Construction of likelihood function
- Maximum likelihood estimation

- Inference
  - Hypothesis testing
  - Confidence interval

Partial likelihood (assuming no tied failure time)

- The likelihood $L(\beta_1, \ldots, \beta_p, \lambda_0) = \prod_i \lambda_{T_i}(\tilde{t}_i)^{\delta_i} S_{T_i}(\tilde{t}_i)$ is not reachable for estimating $\beta_1, \ldots, \beta_p$ with unspecified $\lambda_0(\cdot)$
- Further assumptions
  - K and only K distinct, ordered failure times, say $t_1 < \cdots < t_K$
  - No tied failure time: for each k, there is one and only one individual, say subject i_k, who fails at t_k
  - Risk set $\mathcal{R}(t) = \{i : \widetilde{T}_i \geq t\}$: the set of individuals who are known to survive just prior to time t
- Rephrase $L(\beta_1, \ldots, \beta_p, \lambda_0)$:

$$L(\beta_1, \ldots, \beta_p, \lambda_0) = \prod_{i=1}^{n} \lambda_{T_i}(\tilde{t}_i)^{\delta_i} S_{T_i}(\tilde{t}_i) = \prod_{i=1}^{n} \left\{ \frac{\lambda_{T_i}(\tilde{t}_i)}{\sum_{\ell \in \mathcal{R}(\tilde{t}_i)} \lambda_{T_\ell}(\tilde{t}_i)} \right\}^{\delta_i} \times \left\{ \sum_{\ell \in \mathcal{R}(\tilde{t}_i)} \lambda_{T_\ell}(\tilde{t}_i) \right\}^{\delta_i} \times S_{T_i}(\tilde{t}_i)$$

- Take the partial likelihood (i.e., the first term of the above $L(\boldsymbol{\beta}, \lambda_0)$)

$$pL(\beta_1, \ldots, \beta_p) = \prod_{i=1}^{n} \left\{ \frac{\lambda_{T_i}(\tilde{t}_i)}{\sum_{k \in \mathcal{R}(\tilde{t}_i)} \lambda_{T_k}(\tilde{t}_i)} \right\}^{\delta_i} = \prod_{i=1}^{n} \left\{ \frac{\exp(\sum_{j=1}^{p} x_{ij}\beta_j)}{\sum_{\ell \in \mathcal{R}(\tilde{t}_i)} \exp(\sum_{j=1}^{p} x_{\ell j}\beta_j)} \right\}^{\delta_i} = \prod_{k=1}^{K} \frac{\exp(\sum_{j=1}^{p} x_{i_k j}\beta_j)}{\sum_{\ell \in \mathcal{R}(t_k)} \exp(\sum_{j=1}^{p} x_{\ell j}\beta_j)}$$

  as a surrogate of $L(\beta_1, \ldots, \beta_p, \lambda_0)$ in estimating $\beta_1, \ldots, \beta_p$
  - Cox (1972) argued that $pL(\beta_1, \ldots, \beta_p)$ contained almost all the information about $\beta_1, \ldots, \beta_p$
  - Extensive evidence, both theoretical and numerical, supported this argument in the past few decades
  - $\frac{\exp(\sum_{j=1}^{p} x_{i_k j}\beta_j)}{\sum_{\ell \in \mathcal{R}(t_k)} \exp(\sum_{j=1}^{p} x_{\ell j}\beta_j)}$: the probability of selecting a particular person (here subject i_k) from the risk set at time t_k
- Log-partial likelihood

$$p\ell(\beta_1, \ldots, \beta_p) = \ln pL(\beta_1, \ldots, \beta_p) = \sum_{k=1}^{K} \left\{ \sum_{j=1}^{p} x_{kj}\beta_j - \ln \sum_{\ell \in \mathcal{R}(t_k)} \exp\left(\sum_{j=1}^{p} x_{\ell j}\beta_j \right) \right\}$$

Ex. 5.1 The calculation of partial likelihood

i	$\tilde{t}_i$	δ_i	x_i
1	9	1	4
2	8	0	5
3	6	1	7
4	10	1	3

- Follow the following definition (without reordering failure times) and fill in the table

$$pL(\boldsymbol{\beta}) = \prod_{i=1}^{n} \left\{ \frac{\exp(\sum_{j=1}^{p} x_{ij}\beta_j)}{\sum_{\ell \in \mathcal{R}(\tilde{t}_i)} \exp(\sum_{j=1}^{p} x_{\ell j}\beta_j)} \right\}^{\delta_i}$$

i	$\tilde{t}_i$	δ_i	x_i	$\mathcal{R}(\tilde{t}_i)$	$\left\{\frac{\exp(x_i\beta)}{\sum_{\ell\in\mathcal{R}(\tilde{t}_i)}\exp(x_\ell\beta)}\right\}^{\delta_i}$
1	9	1	4		
2	8	0	5		
3	6	1	7		
4	10	1	3		

Ex. 5.2 The calculation of partial likelihood: comparison of two groups

- Covariate x_i indicating the group label

i	$\tilde{t}_i$	δ_i	x_i	$\mathcal{R}(\tilde{t}_i)$	$\frac{\exp(x_i\beta)}{\sum_{\ell\in\mathcal{R}(\tilde{t}_i)}\exp(x_\ell\beta)}$
1	4	0	0		
2	7	1	0		
3	8	0	0		
4	9	1	0		
5	10	0	0		
6	3	1	1		
7	5	1	1		
8	5	0	1		
9	6	1	1		
10	8	0	1		

```r
library(survival)
data = data.frame(
  tte = c(4,7,8,9,10,3,5,5,6,8),
  delta = c(0,1,0,1,0,1,1,0,1,0),
  x = c(0,0,0,0,0,1,1,1,1,1)
)
fit = coxph(Surv(tte,delta)~x, data = data)
summary(fit)
```

- $\exp(\beta_j)$ is the HR associated with the jump from $group = 0$ to $group = 1$, fixing everything else
- Is there any difference between the survival of the two groups? There are at least four p-values. Which one shall we refer to?
- What are meanings of other digits in the output?

Ex. 5.3. Leukemia data (with tied event/failure times)

Dataset `survival::leukemia` is about survival in patients with Acute Myelogenous Leukemia.

```r
head(survival::leukemia)
```

Partial likelihood (Cox's modification)

- Assumptions
  - K and only K distinct, ordered failure times, say $t_1 < \cdots < t_K$
  - d_k failures at time t_k: there are d_k individuals, say subject $i_{k,1}, \ldots, i_{k,d_k}$, who fail at t_k

- Accordingly

$$pL(\boldsymbol{\beta}) = \prod_{k=1}^{K} \frac{\prod_{i\in\{i_{k,1},\ldots,i_{k,d_k}\}} \exp(\sum_j x_{ij}\beta_j)}{\sum_{D(d_k)\subset\mathcal{R}(t_k)} \prod_{i\in D(d_k)} \exp(\sum_j x_{ij}\beta_j)}$$

  - $D(d_k) \subset \mathcal{R}(t_k)$: a subset of $\mathcal{R}(t_k)$ containing d_k subjects, i.e., a set of d_k individuals who are at risk at t_k
- Labeled as `exact` by `survival::coxph`

Partial likelihood (Breslow's approximation)

- Keeping the assumptions for the Cox's modification
- Substituting $\{\sum_{\ell\in\mathcal{R}(t_k)} \exp(\sum_{j=1}^{p} x_{\ell j}\beta_j)\}^{d_k}$ for the denominator of the Cox's modification

$$pL(\boldsymbol{\beta}) \approx \prod_{k=1}^{K} \frac{\prod_{i\in\{i_{k,1},\ldots,i_{k,d_k}\}} \exp(\sum_j x_{ij}\beta_j)}{\{\sum_{\ell\in\mathcal{R}(t_k)} \exp(\sum_{j=1}^{p} x_{\ell j}\beta_j)\}^{d_k}}$$

- Default tie-handling method in SAS

Partial likelihood (Efron's approximation)

- Keeping the assumptions for the Cox's modification
- Substitute $\prod_{m=1}^{d_k}\{\sum_{\ell\in\mathcal{R}(t_k)} \exp(\sum_{j=1}^{p} x_{\ell j}\beta_j) - \frac{m-1}{d_k}\sum_{i\in\{i_{k,1},\ldots,i_{k,d_k}\}} \exp(\sum_j x_{ij}\beta_j)\}$ for the denominator of Cox's modification

$$pL(\boldsymbol{\beta}) \approx \prod_{k=1}^{K} \frac{\prod_{i\in\{i_{k,1},\ldots,i_{k,d_k}\}} \exp(\sum_j x_{ij}\beta_j)}{\prod_{m=1}^{d_k}\{\sum_{\ell\in\mathcal{R}(t_k)} \exp(\sum_{j=1}^{p} x_{\ell j}\beta_j) - \frac{m-1}{d_k}\sum_{i\in\{i_{k,1},\ldots,i_{k,d_k}\}} \exp(\sum_j x_{ij}\beta_j)\}}$$

- Default tie-handling method by `survival::coxph`

Summary of handling ties

- With no ties, all approximation options give exactly the same results
- With only a few ties, all approximations yield pretty much the same results
- With many ties (relative to the number at risk), both of Breslow's and Efron's approximations yield coefficients that are biased but the Efron's approximation almost always works better than the Breslow's method.
- Computing time of Cox's method is substantially longer than that of approximate methods. But it is not a big issue with today's hardwares.

Revisit Ex. 5.3. Leukemia data (with tied event/failure times)

```r
library(survival)
data = survival::leukemia
fit1 = coxph(Surv(time,status)~x, data = data)
fit2 = coxph(Surv(time,status)~x, data = data, ties = 'efron')
fit3 = coxph(Surv(time,status)~x, data = data, ties = 'breslow')
fit4 = coxph(Surv(time,status)~x, data = data, ties = 'exact')
c(coef(fit1), coef(fit2), coef(fit3), coef(fit4))
```

CIs and hypothesis tests for HRs

- Suppose the HR of interest is the one associated with the one-unit increase of the jth covairate, i.e., $\exp(\beta_j)$
- $\text{var}\{\exp(\hat{\beta}_j)\} \approx \exp(2\hat{\beta}_j)\text{var}(\hat{\beta}_j)$ (delta method)
  - Hence $\text{sd}(\exp(\hat{\beta}_j)) \approx \exp(\hat{\beta}_j)\text{sd}(\hat{\beta}_j)$
- 95% CI for $\exp(\beta_j)$
  - $\exp(\hat{\beta}_j) \pm \Phi^{-1}(.975) \times \text{sd}(\exp(\hat{\beta}_j))$
    - $\Phi^{-1}(.975)$ (≈ 1.96): the .975 quantile of $N(0,1)$
  - $\exp(\hat{\beta}_j \pm \Phi^{-1}(.975) \times \text{sd}(\hat{\beta}_j))$ (preferred; why?)
  - Which one is employed by `survival::coxph`?
- Hypothesis test for $H_0 : \exp(\beta_j) = 1$ (i.e., $\beta_j = 0$) vs. H_1 : otherwise.
  - Wald test statistic: $\hat{\beta}_j/\text{sd}(\hat{\beta}_j) \approx N(0,1)$ under H_0
    - Equivalent to checking whether $\exp(\hat{\beta}_j \pm \Phi^{-1}(.975) \times \text{sd}(\hat{\beta}_j))$ covers 1
- LRT to compare two nested models
  - Model 1 nested to Model 2
    - Model 1: $\lambda(t \mid x_{i1}, \ldots, x_{ip}) = \lambda_0(t)\exp(\sum_{j=1}^{p} x_{ij}\beta_j)$
    - Model 2: $\lambda(t \mid x_{i1}, \ldots, x_{ip}, x_{i,q+1}, \ldots, x_{i,p+q}) = \lambda_0(t)\exp(\sum_{j=1}^{p+q} x_{ij}\beta_j)$
  - H_0 : Model 1 is correct (i.e., $\beta_{p+1} = \cdots = \beta_{p+q} = 0$) vs. H_1 : Model 2 is correct
  - Test statistic: $2(\ln L_{\text{Model2}} - \ln L_{\text{Model1}}) \approx \chi^2(q)$ under H_0

Ex. 5.4. Nursing home data

- Variables:
  - ID: Patient ID
  - lstay: Length of stay of a resident (in days)
  - age: Age of a resident
  - trt: Nursing home assignment (1: receive treatment, 0: control)
  - gender: Gender (1:male, 0:female)
  - marstat: Marital status (1: married, 0: not married)
  - hlstat: Health status (2: second best, 5: worst)
  - cens: Censoring indicator (1:censored, 0: discharged)

```r
options(digits=4)
library(survival)
data = read.csv("NursingHome.csv")
data$event <- 1-data$cens
data$trt = factor(data$trt) # not necessary because it is of two levels
data$gender = factor(data$gender) # not necessary because it is of two levels
data$marstat = factor(data$marstat) # not necessary because it is of two levels
data$hlstat = factor(data$hlstat) # necessary because it is of more than two levels

fit1 <- coxph(
  Surv(lstay,event) ~ trt + age + gender + marstat + hlstat,
  data=data,
  ties = 'exact'
)
summary(fit1)

# Testing if trt is necessary against the full model
```

```r
fit2 <- coxph(
  Surv(lstay,event) ~ age + gender + marstat + hlstat,
  data=data,
  ties = 'exact')
anova(fit1, fit2)

# Testing if trt, age and marstat are necessary against the full model
fit3 <- coxph(
  Surv(lstay,event) ~ gender + hlstat,
  data=data,
  ties = 'exact')
anova(fit1, fit3)

# Testing the necessity of interactions
fit4 <- coxph(
  Surv(lstay,event) ~ age * gender * marstat * hlstat,
  data=data,
  ties = 'exact'
)
anova(fit4, fit3)
```

Baseline cummulative hazard and survival function

- Breslow estimator of $\Lambda_0(t)$ (Breslow, 1972)

$$\hat{\Lambda}_0(t) = \sum_{k:t_k \le t} \frac{d_k}{\sum_{\ell \in \mathcal{R}(t_k)} \exp(\sum_{j=1}^p x_{\ell j}\hat{\beta}_j)}$$

  - d_k: # of events at t_k
  - $\mathcal{R}(t_k)$: the risk set at t_k
  - Reducing to the NA estimator (see Lecture Note Part 2) if all covariates are zeros
- From the perspective of MLE: maximizing the likelihood $L(\beta, \lambda_0)$ instead of the partial likelihood $pL(\beta)$
  - Assuming the cumulative baseline hazard $\Lambda_0(\cdot)$ as piecewise constant between failure times
  - Johansen (1983) proved that
    * $L(\beta, \lambda_0)$ and $pL(\beta)$ share the identical maximizer, say $\hat{\beta}$, with respect to β
    * The maximizer of $L(\beta, \lambda_0)$ with respect to λ_0, say $\hat{\lambda}_0$, satisfies the Breslow estimator
- $\widehat{S}_{T_i}(t) = \exp\{-\hat{\Lambda}_0(t)\}^{\exp(\sum_{j=1}^p x_{ij}\hat{\beta}_j)} = \widehat{S}_0(t)^{\exp(\sum_{j=1}^p x_{ij}\hat{\beta}_j)}$
  - $\widehat{S}_0(t) = \exp\{-\hat{\Lambda}_0(t)\}$: estimated baseline survival function

Ex. 5.4. Revisit the nursing home data

```r
options(digits=4)
library(survival)
data.ex54 = read.csv("NursingHome.csv")
data.ex54$event <- 1-data.ex54$cens
data.ex54$marstat = factor(data.ex54$marstat) # not necessary because it is of two levels
data.ex54$hlstat = factor(data.ex54$hlstat) # necessary because it is of more than two levels
fit.ex54 <- coxph(Surv(lstay,event) ~ marstat + hlstat,  data=data.ex54)
## P.S. note the mandatory standardization of continuous covariates in `survival::coxph`;
## may stop the standardization by changing value of `noncenter`

# baseline hazard and baseline survival
baseline <- basehaz(fit.ex54, centered = FALSE)
```

```r
names(baseline)[1] = 'cum.haz'
baseline$surv = exp(-baseline$cum.haz)
baseline

# Plot the survival function with given values of covariates
newdata.ex54 <- data.frame(
  marstat = factor(c(0,0,1,1)),
  hlstat = factor(c(2,5,2,5))
)
newdata.ex54
cox.predicted.survival <- survfit(fit.ex54, newdata=newdata.ex54)
plot(
  cox.predicted.survival, lty=1:4, col=1:4, lwd=2,
  xlab="Survival Time", ylab="Estimated Survival Probability"
)
legend(
  "topright",
  c(
    "Not married, health status second best",
    "Not married, health status worst",
    "Married, health status second best",
    "Married, health status worst"
  ),
  lty=1:4, col=1:4, lwd=2
)
```